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```
eshaal@DESKTOP-F366D0H: ~/OS-Fall26$ cat user/sleep.c
#include "kernel/types.h"
#include "user/user.h"

int
main(int argc, char *argv[])
{
    if (argc != 2) {
        fprintf(2, "usage: sleep ticks\n");
        exit(1);
    }
    pause(atol(argv[1]));
    exit(0);
}

eshaal@DESKTOP-F366D0H:~/OS-Fall26$ nano Makefile
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ make qemu
mv -f fs.img fs.img.bk
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/eshaal/OS-Fall26-. -DSOL_UTIL
-DLAB_UTIL -march=rvv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen
-fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -
fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno
-pie -no-pie -c -o user/sleep.o user/sleep.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user_sleep user/sleep.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-linux-gnu-objdump -S user/sleep > user/sleep.asm
riscv64-linux-gnu-objdump -t user/sleep | sed '1,/SYMBOL TABLE/d; s/ .* //; /^$/d' > user/sleep.sym
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir
user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sync user/_sleep user
/_mемdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1025 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=non
e,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ sleep 10
$ QEMU: Terminated
```

```
eshaal@DESKTOP-F366D0H: ~/OS-Fall26$ git push origin ProblemSet1
Username for 'https://github.com': EG-05
Password for 'https://EG-05@github.com':
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 573 bytes | 191.00 KiB/s, done.
Total 5 (delta 3), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To https://github.com/EG-05/OS-Fall26.git
  8138d07..2dc3d77 ProblemSet1 -> ProblemSet1
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ git log origin/ProblemSet1 --oneline -3
2dc3d77 (HEAD -> ProblemSet1, origin/ProblemSet1, util, ProblemSet1) Implement sleep
8138d07 (origin/util, origin/HEAD, origin/Assignment1) better test naming
778f26d support git worktree
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ make GRADEFLAGS=sleep grade
make clean
make[1]: Entering directory '/home/eshaal/OS-Fall26'
rm -rf *.tex *.dvi *.idx *.aux *.log *.ind *.ilg *.dSYM *.zip *.pcap \
*/*.o */*.d */*.asm */*.sym \
kernel/kernel fs.img \
mkfs/mkfs _gdbinit \
user/usys.S \
```

```
make[1]: Leaving directory '/home/eshaal/OS-Fall26'
== Test sleep, no arguments == sleep, no arguments: OK (5.2s)
== Test sleep, returns == sleep, returns: OK (1.0s)
== Test sleep, makes syscall == sleep, makes syscall: OK (0.9s)
== Test sleep, zero and non-numeric args return == sleep, zero and non-numeric args return: OK (0.9s)
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ make fmt
git diff
clang-format -i kernel/bio.c kernel/buf.h kernel/console.c kernel/defs.h kernel/elf.h kernel/exec.c kernel/fcntl.h kernel/file.c kernel/file.h kernel/fs.c k
ernel/fs.h kernel/kalloc.c kernel/log.c kernel/main.c kernel/memlayout.h kernel/param.h kernel/pipe.c kernel/plic.c kernel/printk.c kernel/proc.c kernel/pro
c.h kernel/riscv.h kernel/sleeplock.c kernel/sleeplock.h kernel/spinlock.c kernel/spinlock.h kernel/start.c kernel/stat.h kernel/string.c kernel/syscall.c k
ernel/syscall.h kernel/sysfile.c kernel/sysproc.c kernel/trap.c kernel/types.h kernel/uart.c kernel/virtio.c kernel/virtio_disk.c kernel/vm.c kernel/vm.h us
er/_cat.c user/_dorphan.c user/_echo.c user/_forktest.c user/_forphan.c user/_grep.c user/_grind.c user/_init.c user/_kill.c user/_ln.c user/_logstress.c user/_ls.c u
ser/_mемdump.c user/_mkdir.c user/_printf.c user/_rm.c user/_sh.c user/_sleep.c user/_stressfs.c user/_sync.c user/_ulib.c user/_umalloc.c user/_user.h user/_usertests.c
user/_wc.c user/_zombie.c mkfs/mkfs.c
make: clang-format: No such file or directory
make: *** [Makefile:412: fmt] Error 127
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ git add -A
git commit -m "Fix formatting for sleep.c"
git push origin ProblemSet1
On branch ProblemSet1
nothing to commit, working tree clean
Username for 'https://github.com': EG-05
Password for 'https://EG-05@github.com':
Everything up-to-date
```

```
EOF
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ cat user/sixfive.c
make fnt
make qemu
#include "kernel/types.h"
#include "user/user.h"

char *seps = " \r\t\n./,";

void
process(int fd)
{
    char c;
    long num = 0;
    int have = 0;

    while (read(fd, &c, 1) == 1) {
        if (c >= '0' && c <= '9') {
            num = num * 10 + (c - '0');
            have = 1;
        } else if (strchr(seps, c)) {
            if (have && (num % 5 == 0 || num % 6 == 0))
                printf("%ld\n", num);
            num = 0;
            have = 0;
        }
    }
    if (have && (num % 5 == 0 || num % 6 == 0))
        printf("%ld\n", num);
}

int
main(int argc, char *argv[])
{
    if (argc == 1) {
        process(0);
    } else {
        for (int i = 1; i < argc; i++) {
            int fd = open(argv[i], 0);
            if (fd < 0) {
                printf("sixfive: cannot open %s\n", argv[i]);
            }
        }
    }
}
```

```
ne,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ QEMU: Terminated
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ git status
On branch ProblemSet1
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   Makefile

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        user/sixfive.c

no changes added to commit (use "git add" and/or "git commit -a")
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ git add -A
git commit -m "Implement sixfive"
git push origin ProblemSet1
[ProblemSet1 c7fb6c5] Implement sixfive
 2 files changed, 46 insertions(+), 2 deletions(-)
 create mode 100644 user/sixfive.c
Username for 'https://github.com': EG-05
Password for 'https://EG-05@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 811 bytes | 811.00 KiB/s, done.
Total 5 (delta 3), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To https://github.com/EG-05/OS-Fall26.git
 67a3210..c7fb6c5 ProblemSet1 -> ProblemSet1
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ git log origin/ProblemSet1 --oneline -3
c7fb6c5 (HEAD -> ProblemSet1, origin/ProblemSet1) Implement sixfive
67a3210 Ignore llvm installer script
b2e7100 Remove llvm installer script, not part of assignment
```

```
eshaal@DESKTOP-F366D0H: ~ - x
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ cat user/memdump.c
#include "kernel/types.h"
#include "user/user.h"

void
memdump(char *fmt, char *data, int len)
{
    int pos = 0;
    for (int i = 0; fmt[i]; i++) {
        char f = fmt[i];
        int need = f=='i'?4 : f=='p'?8 : f=='h'?2 : f=='c'?1 : f=='s'?8 : 0;

        if (f != 'S' && pos + need > len) {
            printf("memdump: not enough data for '%c'\n", f);
            return;
        }

        if (f == 'i') { int v; memmove(&v, data+pos, 4); printf("%d\n", v); pos += 4; }
        else if (f == 'p') { uint64 v; memmove(&v, data+pos, 8); printf("%lx\n", v); pos += 8; }
        else if (f == 'h') { short v; memmove(&v, data+pos, 2); printf("%d\n", v); pos += 2; }
        else if (f == 'c') { printf("%c\n", data[pos]); pos += 1; }
        else if (f == 's') { char *p; memmove(&p, data+pos, 8); printf("%s\n", p); pos += 8; }
        else if (f == 'S') {
            while (pos < len && data[pos]) printf("%c", data[pos++]);
            printf("\n");
        }
    }
}

eshaal@DESKTOP-F366D0H:~/OS-Fall26$ make fmt
make qemu
clang-format -i kernel/bio.c kernel/buf.h kernel/console.c kernel/defs.h kernel/elf.h kernel/exec.c kernel/fcntl.h kernel/file.c kernel/file.h kernel/fs.c k
kernel/fs.h kernel/kalloc.c kernel/log.c kernel/main.c kernel/memlayout.h kernel/param.h kernel/pipe.c kernel/plic.c kernel/printk.c kernel/proc.c kernel/pro
c.h kernel/riscv.h kernel/sleeplock.c kernel/sleeplock.h kernel/spinlock.c kernel/spinlock.h kernel/start.c kernel/stat.h kernel/string.c kernel/syscall.c k
kernel/syscall.h kernel/sysfile.c kernel/sysproc.c kernel/trap.c kernel/types.h kernel/uart.c kernel/virtio.h kernel/virtio_disk.c kernel/vm.c kernel/vm.h us
er/cat.c user/dorphan.c user/echo.c user/forktest.c user/forphan.c user/grep.c user/grind.c user/init.c user/kill.c user/ln.c user/logstress.c user/lsc.c use
r/memdump.c user/mkdir.c user/printk.c user/rm.c user/sh.c user/sixfive.c user/sleep.c user/stressfs.c user/sync.c user/ulib.c user/umalloc.c user/user.h us
er/usertests.c user/wc.c user/zombie.c mkfs/mkfs.c
mv -f fs.img fs.img.bk
mv: cannot stat 'fs.img': No such file or directory
make: [Makefile:292: newfs.img] Error 1 (ignored)
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/eshaal/OS-Fall26. -DSOL_UTIL

eshaal@DESKTOP-F366D0H: ~ - x
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ cat > user/memdump.c << 'EOF'
#include "kernel/types.h"
#include "user/user.h"
#include "kernel/fcntl.h"

void memdump(char *fmt, char *data, int len);

int
main(int argc, char *argv[])
{
    if (argc == 1) {
        printf("Example 1:\n");
        int a[2] = {61810, 2026};
        memdump("ii", (char *)a, sizeof(a));

        printf("Example 2:\n");
        memdump("S", "a string", sizeof("a string"));

        printf("Example 3:\n");
        char ss = "another";
        memdump("s", (char *)&s, sizeof(s));

        struct sss {
            char *ptr;
            int num1;
            short num2;
            char byte;
            char bytes[8];
        } example;

        example.ptr = "hello";
        example.num1 = 1819438967;
        example.num2 = 100;
        example.byte = 'z';
        strcpy(example.bytes, "xyzyzy");

        printf("Example 4:\n");
        memdump("pihcS", (char *)&example, sizeof(example));

        printf("Example 5:\n");
        memdump("scccc", (char *)&example, sizeof(example));
    }
}
```

```
eshaal@DESKTOP-F366D0H: ~/OS-Fall26$ git add -A
git commit -m "Implement memdump"
git push origin ProblemSet1
[ProblemSet1 83963b9] Implement memdump
1 file changed, 48 insertions(+), 43 deletions(-)
Username for 'https://github.com': EG-05
Password for 'https://github.com': 
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 1.08 KiB | 1.08 MiB/s, done.
Total 4 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/EG-05/OS-Fall26.git
  4812c8a..83963b9 ProblemSet1 -> ProblemSet1
eshaal@DESKTOP-F366D0H: ~/OS-Fall26$ git log origin/ProblemSet1 --oneline -3
83963b9 (HEAD -> ProblemSet1, origin/ProblemSet1) Implement memdump
4812c8a Updated memdump.c
c7fb6c5 Implement sixfive
eshaal@DESKTOP-F366D0H: ~/OS-Fall26$ cat>user/find.c<<'EOF'
> #include "kernel/types.h"
#include "kernel/stat.h"
#include "kernel/fs.h"
#include "user/user.h"

void
find(char *path, char *name)
{
    char buf[512], *p;
    int fd;
    struct dirent de;
    struct stat st;

    if ((fd = open(path, 0)) < 0) {
        fprintf(2, "find: cannot open %s\n", path);
        return;
    }
    if (fstat(fd, &st) < 0) {
        fprintf(2, "find: cannot stat %s\n", path);
        close(fd);
    }

make[1]: Leaving directory '/home/eshaal/OS-Fall26'
== Test find, in current directory == find, in current directory: OK (4.2s)
  (Old xv6.out.find_curdir failure log removed)
== Test find, in sub-directory == find, in sub-directory: OK (1.2s)
  (Old xv6.out.find_subdir failure log removed)
== Test find, recursive == find, recursive: OK (1.4s)
  (Old xv6.out.find_recursive failure log removed)
== Test find, missing target, no duplicates, directory match == find, missing target, no duplicates, directory match: FAIL (1.6s)
  times directory dGyxxXta reported
  got:
    0
  expected:
    1
  QEMU output saved to xv6.out.find_corner
make: *** [Makefile:369: grade] Error 1
eshaal@DESKTOP-F366D0H: ~/OS-Fall26$ nano user/find.c
eshaal@DESKTOP-F366D0H: ~/OS-Fall26$ make fmt
make GRADEFLAGS=find grade
clang-format -i kernel/bio.c kernel/buf.h kernel/console.c kernel/defs.h kernel/elf.h kernel/errno.c kernel/fcntl.h kernel/file.c kernel/file.h kernel/fs.c k
ernel/fs.h kernel/kalloc.c kernel/log.c kernel/main.c kernel/memlayout.h kernel/param.h kernel/pipe.c kernel/plie.c kernel/printk.c kernel/proc.c kernel/pno
c.h kernel/riscv.h kernel/sleeplock.c kernel/sleeplock.h kernel/spinlock.c kernel/spinlock.h kernel/start.c kernel/stat.h kernel/string.c kernel/syscall.c k
ernel/syscall.h kernel/sysfile.c kernel/sysproc.c kernel/trap.c kernel/types.h kernel/uart.c kernel/virtio.h kernel/virtio_disk.c kernel/vm.c kernel/vm.h us
er/cat.c user/dorphan.c user/echo.c user/find.c user/forktest.c user/forphan.c user/grep.c user/grind.c user/init.c user/kill.c user/ln.c user/logstress.c u
ser/ls.c user/memdump.c user/mkdir.c user/printf.c user/rm.c user/sh.c user/sixfive.c user/sleep.c user/stressfs.c user/sync.c user/ulib.c user/umalloc.c us
er/user.h user/usertests.c user/wc.c user/zombie.c mkfs/mkfs.c
make clean
make[1]: Entering directory '/home/eshaal/OS-Fall26'
rm -rf *.tex *.dvi *.idx *.aux *.log *.ind *.ilg *.dSYM *.zip *.pcap \
*/*.o */*.d */*.asm */*.sym \
kernel/kernel fs.img \
mkfs/mkfs_gdbinit \
user/usys.S \
user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_gr
ind user/_wc user/_zombie user/_logstress user/_forphan user/_sync user/_sleep user/_sixfive user/_find user/_memdump
make[1]: Leaving directory '/home/eshaal/OS-Fall26'
./grade-lab-util find
make[1]: Entering directory '/home/eshaal/OS-Fall26'
riscv64-linux-gnu-gcc -march=rv64gc -g -ffile-prefix-map=/home/eshaal/OS-Fall26=. -c -o kernel/entry.o kernel/entry.S
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-gomit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/eshaal/OS-Fall26=. -D$OL_UTIL
-DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen
-fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -
```

```
eshaal@DESKTOP-F366D0H: ~
kernel/uart.o kernel/spinlock.o
riscv64-linux-gnu-ld: warning: kernel/kernel has a LOAD segment with RWX permissions
riscv64-linux-gnu-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-linux-gnu-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* //; /^$/d' > kernel/kernel.sym
make[1]: Leaving directory '/home/eshaal/OS-Fall26'
== Test find, in current directory == find, in current directory: OK (4.1s)
== Test find, in sub-directory == find, in sub-directory: OK (1.0s)
== Test find, recursive == find, recursive: OK (1.3s)
== Test find, missing target, no duplicates, directory match == find, missing target, no duplicates, directory match: OK (1.5s)
(Old xv6.out.find_corner failure log removed)
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ git add -A
git commit -m "Implement find"
git push origin ProblemSet1
[ProblemSet1 c141a57] Implement find
2 files changed, 63 insertions(+)
create mode 100644 user/find.c
Username for 'https://github.com': EG-05
Password for 'https://EG-05@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/EG-05/OS-Fall26.git/'
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ git add -A
git commit -m "Implement find"
git push origin ProblemSet1
On branch ProblemSet1
nothing to commit, working tree clean
Username for 'https://github.com': EG-05
Password for 'https://EG-05@github.com':
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 949 bytes | 474.00 KiB/s, done.
Total 5 (delta 3), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To https://github.com/EG-05/OS-Fall26.git
03963b9..c141a57 ProblemSet1 -> ProblemSet1
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ git log origin/ProblemSet1 --oneline -3
c141a57 (HEAD -> ProblemSet1, origin/ProblemSet1) Implement find
03963b9 Implement memdump
4812c8a Updated memdump.c
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ make fmt

#include "kernel/param.h"
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ cat > user/find.c << 'EOF'
#include "kernel/types.h"
#include "kernel/stat.h"
#include "kernel/fs.h"
#include "kernel/param.h"
#include "user/user.h"

void
runcmd(char **cmdv, int cmdc, char *file)
{
    char *args[MAXARG];
    int i;

    for (i = 0; i < cmdc; i++)
        args[i] = cmdv[i];
    args[i++] = file;
    args[i] = 0;

    int pid = fork();
    if (pid < 0) {
        fprintf(2, "find: fork failed\n");
        return;
    }
    if (pid == 0) {
        exec(args[0], args);
        fprintf(2, "find: exec %s failed\n", args[0]);
        exit(1);
    }
    wait(0);
    EOFxit(0);v[1], argv[2], execv, execc)[-exec cmd ...]\n");) == 0)
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ head -5 user/find.c
#include "kernel/types.h"
#include "kernel/stat.h"
#include "kernel/fs.h"
#include "kernel/param.h"
#include "user/user.h"

make qemu
clang-format -i kernel/bio.c kernel/buf.h kernel/console.c kernel/defs.h kernel/elf.h kernel/exec.c kernel/fcntl.h kernel/file.c kernel/file.h kernel/fs.c k
ernel/fs.h kernel/kalloc.c kernel/log.c kernel/main.c kernel/memlayout.h kernel/param.h kernel/pipe.c kernel/plic.c kernel/printk.c kernel/proc.c kernel/pro
```

```
eshaal@DESKTOP-F366D0H: ~ - v
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ git add -A
git commit -m "Editing find for exec"
git push origin ProblemSet1
[ProblemSet1 8c8ab63] Editing find for exec
1 file changed, 50 insertions(+), 7 deletions(-)
Username for 'https://github.com': EG-05
Password for 'https://github.com':
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 785 bytes | 392.00 KiB/s, done.
Total 4 (delta 3), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To https://github.com/EG-05/OS-Fall26.git
   c141a57..8c8ab63 ProblemSet1 -> ProblemSet1
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ git status
On branch ProblemSet1
nothing to commit, working tree clean
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ git log origin/ProblemSet1 --oneline -3
8c8ab63 (HEAD -> ProblemSet1, origin/ProblemSet1) Editing find for exec
c141a57 Implement find
03963b9 Implement memdump
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ cat > time.txt
5
EOF
^[[A^X'eof'
^X^X^Z
[1]+  Stopped                  cat > time.txt
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ cat user/time
cat: user/time: No such file or directory
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ cat > user/time.txt << 'EOF'
> 5
> EOF
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ make fmt
make qemu
clang-format -i kernel/bio.c kernel/buf.h kernel/console.c kernel/defs.h kernel/elf.h kernel/exec.c kernel/fcntl.h kernel/file.c kernel/file.h kernel/fs.c k
kernel/fs.h kernel/kalloc.c kernel/log.c kernel/main.c kernel/memlayout.h kernel/param.h kernel/pipe.c kernel/plic.c kernel/printk.c kernel/proc.c kernel/pro
c.h kernel/riscv.h kernel/sleeplock.c kernel/sleeplock.h kernel/spinlock.c kernel/spinlock.h kernel/start.c kernel/stat.h kernel/string.c kernel/syscall.c k
kernel/syscall.h kernel/sysfile.c kernel/sysproc.c kernel/trap.c kernel/types.h kernel/uart.c kernel/virtio.h kernel/virtio_disk.c kernel/vm.c kernel/vm.h us
er/cat.c user/dorphan.c user/echo.c user/find.c user/forktest.c user/forphan.c user/grep.c user/grind.c user/init.c user/kill.c user/ln.c user/logstress.c u
er/cat.c user/dorphan.c user/echo.c user/find.c user/forktest.c user/forphan.c user/grep.c user/grind.c user/init.c user/kill.c user/ln.c user/logstress.c u
32°C Mostly cloudy
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eshaal@DESKTOP-F366D0H: ~ - v
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ nano user/time.txt
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ cat user/time.txt
5
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ make fmt
make qemu
clang-format -i kernel/bio.c kernel/buf.h kernel/console.c kernel/defs.h kernel/elf.h kernel/exec.c kernel/fcntl.h kernel/file.c kernel/file.h kernel/fs.c k
kernel/fs.h kernel/kalloc.c kernel/log.c kernel/main.c kernel/memlayout.h kernel/param.h kernel/pipe.c kernel/plic.c kernel/printk.c kernel/proc.c kernel/pro
c.h kernel/riscv.h kernel/sleeplock.c kernel/sleeplock.h kernel/spinlock.c kernel/spinlock.h kernel/start.c kernel/stat.h kernel/string.c kernel/syscall.c k
kernel/syscall.h kernel/sysfile.c kernel/sysproc.c kernel/trap.c kernel/types.h kernel/uart.c kernel/virtio.h kernel/virtio_disk.c kernel/vm.c kernel/vm.h us
er/cat.c user/dorphan.c user/echo.c user/find.c user/forktest.c user/forphan.c user/grep.c user/grind.c user/init.c user/kill.c user/ln.c user/logstress.c u
ser/ls.c user/memdump.c user/mkdir.c user/print.c user/rw.c user/sh.c user/sixfive.c user/sleep.c user/stressfs.c user/sync.c user/ulib.c user/umalloc.c us
er/user.h user/usertests.c user/wc.c user/zombie.c mkfs/mkfs.c
mv -f fs.img fs.img.bk
mv: cannot stat 'fs.img': No such file or directory
make: [Makefile:293: newfs.img] Error 1 (ignored)
gcc -DSOL_UTIL -DLAB_UTIL -Wno-unknown-attributes -Werror -Wall -I. -o mkfs/mkfs mkfs/mkfs.c
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/eshaal/OS-Fall26- -DSOL_UTIL
-DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen
-fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -
fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno
-pie -no-pie -c -o user/ulib.o user/ulib.c
perl user/usys.pl > user/usys.S
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/eshaal/OS-Fall26- -DSOL_UTIL
-DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen
-fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -
fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno
-pie -no-pie -c -o user/usys.o user/usys.S
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/eshaal/OS-Fall26- -DSOL_UTIL
-DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen
-fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -
fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno
-pie -no-pie -c -o user/printf.o user/printf.c
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/eshaal/OS-Fall26- -DSOL_UTIL
-DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen
-fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -
fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno
-pie -no-pie -c -o user/umalloc.o user/umalloc.c
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/eshaal/OS-Fall26- -DSOL_UTIL
-DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen
-fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -
fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno
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15/09/2026
```

```
eshaal@DESKTOP-F366D0H: ~ - x + v
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ cat > user/uptime.c << 'EOF'
#include "kernel/types.h"
#include "user/user.h"

int
main(void)
{
    printf("%d\n", uptime());
    exit(0);
}
EOF
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ nano Makefile
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ nano Makefile
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ grep -n uptime user/user.h user/usys.pl kernel/sysproc.c kernel/syscall.c kernel/syscall.h
user/user.h:26:int uptime(void);
user/usys.pl:44:entry("uptime");
kernel/sysproc.c:104:sys_uptime(void)
kernel/syscall.c:97:extern uint64 sys_uptime(void);
kernel/syscall.c:124: [SYS_uptime] = sys_uptime,
kernel/syscall.h:15:#define SYS_uptime 14
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ nano Makefile
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ make fnt make qemu
clang-format -i kernel/bio.c kernel/buf.h kernel/console.c kernel/defs.h kernel/elf.h kernel/exec.c kernel/fcntl.h kernel/file.c kernel/file.h kernel/fs.c k
ernel/fs.h kernel/kalloc.c kernel/log.c kernel/main.c kernel/memlayout.h kernel/param.h kernel/pipe.c kernel/plie.c kernel/printk.c kernel/proc.c kernel/pro
c.h kernel/riscv.h kernel/sleeplock.c kernel/sleeplock.h kernel/spinlock.c kernel/start.c kernel/stat.h kernel/string.c kernel/syscall.c k
ernel/syscall.h kernel/sysfile.c kernel/sysproc.c kernel/trap.c kernel/types.h kernel/uart.c kernel/virtio.h kernel/virtio_disk.c kernel/vm.c kernel/vm.h us
er/cat.c user/dorphan.c user/echo.c user/find.c user/forktest.c user/forphan.c user/grep.c user/grind.c user/init.c user/kill.c user/lnc.c user/logstress.c u
ser/ls.c user/memdump.c user/mkdir.c user/printf.c user/rm.c user/sh.c user/sixfive.c user/sleep.c user/stressfs.c user/sync.c user/ulib.c user/umalloc.c us
er/uptime.c user/user.h user/usertests.c user/wc.c user/zombie.c mkfs/mkfs.c
make: *** No rule to make target 'make'. Stop.
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ make qemu
mv -f fs.img fs.img.bk
mv: cannot stat 'fs.img': No such file or directory
make: [Makefile:293: newfs.img] Error 1 (ignored)
gcc -DSOL_UTIL -DLAB_UTIL -Wno-unknown-attributes -Werror -Wall -I. -o mkfs/mkfs mkfs/mkfs.c
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/eshaal/OS-Fall26- -DSOL_UTIL
-DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen
-fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -
fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno
-pie -no-pie -c -o user/ulib.o user/ulib.c
perl user/usys.pl > user/usys.$
-DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen
-fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -
fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno
-pie -no-pie -c -o user/ulib.o user/ulib.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_find user/find.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-linux-gnu-objdump -S user/_find > user/find.asm
riscv64-linux-gnu-objdump -t user/_find | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/find.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/eshaal/OS-Fall26- -DSOL_UTIL
-DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen
-fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -
fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno
-pie -no-pie -c -o user/uptime.o user/uptime.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_uptime user/uptime.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-linux-gnu-objdump -S user/uptime > user/uptime.asm
riscv64-linux-gnu-objdump -t user/uptime | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/uptime.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/eshaal/OS-Fall26- -DSOL_UTIL
-DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen
-fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -
fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno
-pie -no-pie -c -o user/memdump.o user/memdump.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_memdump user/memdump.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-linux-gnu-objdump -S user/memdump > user/memdump.asm
riscv64-linux-gnu-objdump -t user/memdump | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/memdump.sym
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir
user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sync user/_sleep user/
_sixfive user/_find user/_uptime user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1147 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=no
ne,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ uptime
 51
$ QEMU: Terminated
eshaal@DESKTOP-F366D0H:~/OS-Fall26$ |
```

```
eshaal@DESKTOP-F366D0H: ~
c.h kernel/riscv.h kernel/sleeplock.c kernel/sleeplock.h kernel/spinlock.c kernel/spinlock.h kernel/start.c kernel/stat.h kernel/string.c kernel/syscall.c k
erl/syscall.h kernel/sysfile.c kernel/sysproc.c kernel/trap.c kernel/types.h kernel/uart.c kernel/virtio.h kernel/virtio_disk.c kernel/vm.c kernel/vm.h us
er/cat.c user/dorphan.c user/echo.c user/find.c user/forktest.c user/forphan.c user/grep.c user/grind.c user/init.c user/kill.c user/ln.c user/logstress.c u
ser/ls.c user/memdump.c user/mkdir.c user/printf.c user/rm.c user/sh.c user/sixfive.c user/sleep.c user/stressfs.c user/sync.c user/ulib.c user/umalloc.c us
er/uptime.c user/user.h user/usertests.c user/wc.c user/zombie.c mkfs/mkfs.c
mv -f fs.img fs.img.bk
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/eshaal/OS-Fall26- -DSOL_UTIL
-DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen
-fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -
fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno
-pie -no-pie -c -o user/find.o user/find.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_find user/find.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-linux-gnu-objdump -S user/_find > user/find.asm
riscv64-linux-gnu-objdump -t user/_find | sed '1,/SYMBOL TABLE/d; s/ . * / /; /'$d' > user/find.sym
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir
user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sync user/_sleep user/_
sixfive user/_find user/_uptime user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1149 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=no
ne,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ echo >b
$ echo >b2
$ find . "b$"
$ find . b
exec $ failed
$ find . b
./zombie
./b
./b2
./b2
$ find . "b$"
./b
$ |

29°C
Mostly cloudy
1922
15/09/2026

eshaal@DESKTOP-F366D0H: ~
ltn-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/sysfile.o kernel/sysfile.c
riscv64-linux-gnu-gcc -march=rv64gc -g -ffile-prefix-map=/home/eshaal/OS-Fall26- -c -o kernel/kernelvec.o kernel/kernelvec.S
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/eshaal/OS-Fall26- -DSOL_UTIL -DLAB_UTIL -m
arch=rv64gc -std=gnu99 -MD -mmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-bui
ltn-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-bui
ltn-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/plc.o kernel/plc.c
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/eshaal/OS-Fall26- -DSOL_UTIL -DLAB_UTIL -m
arch=rv64gc -std=gnu99 -MD -mmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-bui
ltn-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-bui
ltn-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/virtio_disk.o kernel/virtio_d
isk.c
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/eshaal/OS-Fall26- -DSOL_UTIL -DLAB_UTIL -m
arch=rv64gc -std=gnu99 -MD -mmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-bui
ltn-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-bui
ltn-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/start.o kernel/start.c
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/eshaal/OS-Fall26- -DSOL_UTIL -DLAB_UTIL -m
arch=rv64gc -std=gnu99 -MD -mmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-bui
ltn-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-bui
ltn-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/console.o kernel/console.c
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/eshaal/OS-Fall26- -DSOL_UTIL -DLAB_UTIL -m
arch=rv64gc -std=gnu99 -MD -mmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-bui
ltn-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-bui
ltn-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/spinlock.o kernel/spinlock.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T kernel/kernel.ld -o kernel/kernel kernel/entry.o kernel/kalloc.o kernel/string.o kernel/main.o kernel/vm.o kernel/proc.o ker
nel/swtch.o kernel/trampoline.o kernel/trap.o kernel/syscall.o kernel/sysproc.o kernel/bio.o kernel/fs.o kernel/log.o kernel/sleeplock.o kernel/file.o kernel/pipe.o kerne
l/proc.o kernel/sysfile.o kernel/kernelvec.o kernel/plc.o kernel/virtio_disk.o kernel/start.o kernel/console.o kernel/printk.o kernel/uart.o kernel/spinlock.o
riscv64-linux-gnu-ld: warning: kernel/kernel has a LOAD segment with RWX permissions
riscv64-linux-gnu-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-linux-gnu-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ . * / /; /'$d' > kernel/kernel.sym
make[1]: Leaving directory '/home/eshaal/OS-Fall26'
== Test exec == exec: OK (0.1s)
(Old xv6.out.find_exec failure log removed)
== Test exec, multiple args == exec, multiple args: OK (1.0s)
(Old xv6.out.find_exec_args failure log removed)
== Test exec via sh script == exec via sh script: OK (1.4s)
(Old xv6.out.test_find_sh failure log removed)
eshaal@DESKTOP-F366D0H: ~/OS-Fall26$ |

28°C
Partly cloudy
1951
15/09/2026
```

```
eshaal@DESKTOP-F366DDH: ~  
202 | exit(0);  
|  
user/sh.c:203:1: error: expected identifier or '(' before ')' token  
203 | )  
|  
ccl: note: unrecognized command-line option '-Wno-unknown-attributes' may have been intended to silence earlier diagnostics  
make: *** [<built-in>: user/sh.o] Error 1  
eshaal@DESKTOP-F366DDH:~/OS-Fall26$ /nano user/sh.c  
-bash: /nano: No such file or directory  
eshaal@DESKTOP-F366DDH:~/OS-Fall26$ nano user/sh.c  
eshaal@DESKTOP-F366DDH:~/OS-Fall26$ make fat  
clang-format -i kernel/bio.c kernel/buf.h kernel/console.c kernel/defs.h kernel/elf.h kernel/exec.c kernel/fcntl.h kernel/file.c kernel/file.h kernel/fs.c kernel/fs.h ker  
nel/kalloc.c kernel/log.c kernel/main.c kernel/memlayout.h kernel/param.h kernel/pipe.c kernel/plc.c kernel/printk.c kernel/proc.c kernel/proc.h kernel/riscv.h kernel/st  
eoplock.c kernel/sleeplock.h kernel/spinlock.c kernel/spinlock.h kernel/start.c kernel/stat.h kernel/string.c kernel/syscall.c kernel/syscall.h kernel/sysfile.c kernel/sy  
sproc.c kernel/trap.c kernel/types.h kernel/uart.c kernel/virtio.h kernel/virtio_disk.c kernel/vm.c kernel/vm.h user/cat.c user/dorphan.c user/echo.c user/find.c user/for  
ktest.c user/forphan.c user/grep.c user/grind.c user/init.c user/kill.c user/ln.c user/logstress.c user/ls.c user/memdump.c user/mkdir.c user/printk.c user/rm.c user/sh.c  
user/sixfive.c user/sleep.c user/stressfs.c user/sync.c user/ulib.c user/umalloc.c user/uptime.c user/user.h user/usertests.c user/wc.c user/zombie.c mkfs/mkfs.c  
eshaal@DESKTOP-F366DDH:~/OS-Fall26$ make qemu  
mv -f fs.img fs.img.Dk  
mv: cannot stat 'fs.img': No such file or directory  
make: [Makefile:293: newfs.img] Error 1 (ignored)  
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/eshaal/OS-Fall26-. -DSOL_UTIL -DLAB_UTIL -m  
arch=riscv64 -std=gnu99 -MD -mmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-bui  
ltn-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-bu  
iltin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -fno-builtin-putc -fno-builtin-putc -fno-builtin-putc -fno-builtin-putc -fno-bu  
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_sh user/sh.o user/ulib.o user/usys.o user/printk.o user/umalloc.o  
riscv64-linux-gnu-objdump -S user/_sh > user/sh.asm  
riscv64-linux-gnu-objdump -t user/_sh | sed '1,/SYMBOL TABLE/d; s/ .* //; /$/d' > user/sh.sym  
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user  
/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sync user/_sleep user/_sixfive user/_find user/_up  
time user/_memdump  
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000  
ballocc: first 1150 blocks have been allocated  
ballocc: write bitmap block at sector 46  
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,  
id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0  
  
xv6 kernel is booting  
  
hart 1 starting  
hart 2 starting  
init: starting sh  
$ sleep 50 &  
$ wait  
$ |
```

```
eshaal@DESKTOP-F366DDH: ~  
ballocc: first 1150 blocks have been allocated  
ballocc: write bitmap block at sector 46  
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,  
id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0  
  
xv6 kernel is booting  
  
hart 1 starting  
hart 2 starting  
init: starting sh  
$ echo hello  
hello  
$ echo bg &  
bg  
$ wait  
$ history  
1 echo hello  
2 echo bg &  
3 wait  
4 history  
$ QEMU: Terminated
```

```
GNU nano 7.2 user/sh.c  
}  
}  
  
// Read and run input commands.  
while (getcmd(buf, sizeof(buf)) >= 0) {  
    char *cmd = buf;  
    while (*cmd == ' ' || *cmd == '\t')  
        cmd++;  
    if (*cmd == '\n') // is a blank command  
        continue;  
    if (strcmp(cmd, "wait\n") == 0) {  
        while (wait(0) >= 0)  
            ;  
        continue;  
    }  
    if (strcmp(cmd, "history\n") == 0) {  
        int start = histcount > HISTSIZE ? histcount - HISTSIZE : 0;  
        for (int i = start; i < histcount; i++)  
            fprintf(2, "%d %s", i + 1, hist[i % HISTSIZE]);  
        continue;  
    }  
    if (cmd[0] == 'c' && cmd[1] == 'd' && cmd[2] == ' ') {  
        // Chdir must be called by the parent, not the child.  
        cmd[strlen(cmd) - 1] = 0; // chop \n  
        if (chdir(cmd + 3) < 0)  
            fprintf(2, "cannot cd %s\n", cmd + 3);  
    } else if (strcmp(cmd, "wait\n") == 0) {  
        // Reap all currently outstanding background children.  
        while (wait(0) >= 0)  
            ;  
    } else {  
        if (fork1() == 0)  
            runcmd(parsecmd(cmd));  
        wait(0);  
    }  
}  
exit(0);  
}  
  
void  
panic(char *s)
```

```
GNU nano 7.2 user/sh.c *
// Shell.

#include "kernel/types.h"
#include "user/user.h"

#define HISTSIZE 16
static char hist[HISTSIZE][100];
static int histcount = 0;
static int interactive = 1;

#include "kernel/fcntl.h"
#include "kernel/stat.h"

// Parsed command representation
#define EXEC 1
#define REDIR 2
#define PIPE 3
#define LIST 4
#define BACK 5

#define MAXARGS 10

struct cmd {
    int type;
};

struct execcmd {
    int type;
    char *argv[MAXARGS];
    char *eargv[MAXARGS];
};

struct redircmd {
    int type;
    struct cmd *cmd;
    char *file;
    char *efile;
    int mode;
    int fd;
};

dup(p[0]);
close(p[0]);
close(p[1]);
runcmd(pcmd->right);
}
close(p[0]);
close(p[1]);
wait(0);
wait(0);
break;

case BACK:
    bcmd = (struct backcmd *)cmd;
    if (fork1() == 0)
        runcmd(bcmd->cmd);
    break;
}
exit(0);
}

int
getcmd(char *buf, int nbuf)
{
    if (interactive)
        write(2, "$ ", 2);
    memset(buf, 0, nbuf);
    gets(buf, nbuf);
    if (buf[0] != 0) {
        strcpy(hist[histcount % HISTSIZE], buf);
        histcount++;
    }
    if (buf[0] == 0) // EOF
        return -1;
    return 0;
}

int
main(void)
{
    static char buf[100];
    int fd;
}
```

```
GNU nano 7.2 user/sh.c *
while (*cmd == ' ' || *cmd == '\t')
    cmd++;
if (*cmd == '\n') // is a blank command
    continue;
if (strcmp(cmd, "wait\n") == 0) {
    while (wait(0) >= 0)
        continue;
}
if (strcmp(cmd, "history\n") == 0) {
    int start = histcount > HISTSIZE ? histcount - HISTSIZE : 0;
    for (int i = start; i < histcount; i++)
        fprintf(2, "%d %s", i + 1, hist[i % HISTSIZE]);
    continue;
}
if (cmd[0] == 'c' && cmd[1] == 'd' && cmd[2] == ' ') {
    // Chdir must be called by the parent, not the child.
    cmd[strlen(cmd) - 1] = 0; // chop \n
    if (chdir(cmd + 3) < 0)
        fprintf(2, "cannot cd %s\n", cmd + 3);
} else if (strcmp(cmd, "wait\n") == 0) {
    // Reap all currently outstanding background children.
    while (wait(0) >= 0)
        continue;
} else {
    if (fork1() == 0)
        runcmd(parsecmd(cmd));
    wait(0);
}
}
exit(0);
}

void
panic(char *s)
{
    fprintf(2, "%s\n", s);
    exit(1);
}

int
getcmdline(char *buf, int nbuf)
{
    int i = 0;
    char c;

    while (i < nbuf - 1) {
        if (read(0, &c, 1) != 1)
            return -1;

        if (c == '\n') {
            buf[i] = '\0';
            write(2, "\n", 1);
            return 0;
        }

        if (c == '\t') {
            buf[i] = '\0';

            // Tab completion will go here.

            continue;
        }

        if (c == '\b' || c == 127) {
            if (i > 0) {
                i--;
                write(2, "\b \b", 3);
            }
            continue;
        }

        if (i < nbuf - 1) {
            buf[i++] = c;
            write(2, &c, 1);
        }
    }
    buf[i] = '\0';
    return 0;
}
```

```
eshaal@DESKTOP-F366D0H: ~  
id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0  
  
xv6 kernel is booting  
hart 1 starting  
hart 2 starting  
init: starting sh  
$ history  
history  
1 history  
$ wait  
wait  
$ ls  
ls  
.  
..  
README  
findtest.sh  
sixfive.txt  
cat  
echo  
forktest  
grep  
init  
kill  
ln  
ls  
mkdir  
rm  
sh  
stressfs  
usertests  
grind  
wc  
zombie  
logstress  
forphan  
dorphan  
sync  
sleep  
sixfive  
find  
uptime  
memdump  
console  
1 1 1024  
1 1 1024  
2 2 2441  
2 3 91  
2 4 19  
2 5 35376  
2 6 34312  
2 7 17280  
2 8 42736  
2 9 34768  
2 10 34240  
2 11 34032  
2 12 41520  
2 13 34280  
2 14 34264  
2 15 68360  
2 16 35160  
2 17 199208  
2 18 58504  
2 19 36336  
2 20 33648  
2 21 36104  
2 22 35848  
2 23 34488  
2 24 33712  
2 25 34032  
2 26 35864  
2 27 45520  
2 28 33656  
2 29 42000  
3 30 0  
  
Missing file for redirection QEMU: terminated  
eshaal@DESKTOP-F366D0H: ~/OS-Fall26$ make qemu  
mv -f fs.img fs.img.bk  
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user  
/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sync user/_sleep user/_sixfive user/_find user/_up  
time user/_memdump  
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000  
ballocc: first 1153 blocks have been allocated  
ballocc: write bitmap block at sector 46  
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,  
id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0  
  
xv6 kernel is booting  
hart 1 starting  
hart 2 starting  
init: starting sh  
$ ec  
echo  
$ |
```